Synthetic Market Phase 3 Coupled Geometry

Coupled-factor geometry results for the counting-manifolds-inspired market-state program. This report moves past the single-factor search and asks whether small interacting factor sets form cleaner low-dimensional structure than the scalar sweeps from Phases 1 and 2.

DATE	PROMPTS	FAMILIES	STAGE
20 March 2026	1,089 market-only	726 dense, 363 minimal	Coupled geometry

MAIN READ

Phase 3 supports a coupled-space view of market representation, but it does not replace the scalar story.

The strongest pair, momentum × flow, forms a reproducible late-layer geometry that survives within-template checks almost unchanged (dense 0.708, minimal 0.693). But it still does not beat the best isolated scalar baseline from Phase 2 (`pct_5m` at 0.741). The result favors a layered picture: strong single-factor candidates plus a smaller number of modestly ordered coupled spaces, not one dominant universal manifold.

PHASE 3 SCALE	BEST COUPLED GEOM.	BEST REGRESSION R^2	PHASE 2 `pct_5m`
1,089	**0.708**	**0.998**	**0.741**
market-only prompts	momentum × flow, dense L28	dense risk/edge, L4	best scalar baseline

Scope

Phase 3 is the first explicit Stage 2 experiment from the updated counting-manifolds plan: hold onto the scalar search, but stop expecting the whole market to collapse into one universal one-dimensional variable.

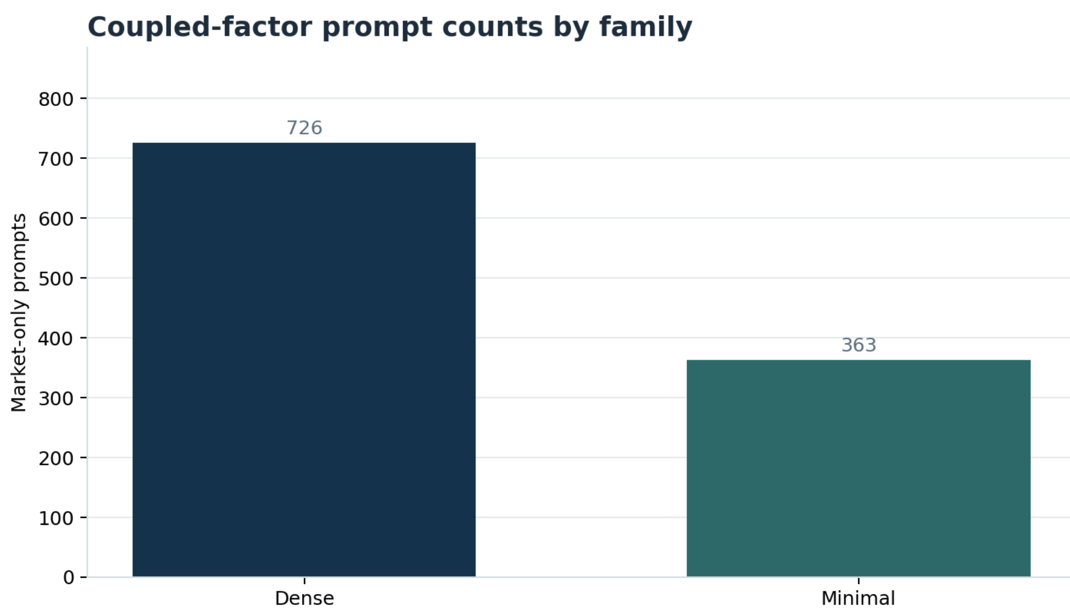
- **Stage 1 recap.** Phases 1 and 2 showed that some synthetic scalar variables, especially `pct_5m`, have meaningful low-dimensional order, but not a universal 1D manifold.
- **Stage 2 question.** Do naturally interacting factor pairs produce cleaner low-dimensional geometry than the isolated scalar sweeps?
- **Controlled synthetic setting.** The prompts remain market-only, with neutral assets `A/B/C/D`, clean by-construction labels, and the same pooled row-state analysis path as earlier phases.

The coupled families in this phase are:

- `pct_5m × unique_traders_5m` as a momentum × participation candidate.
- `pct_5m × top20_holder_pct` as a momentum × concentration candidate.
- `pct_5m × net_flow_5m` as a momentum × flow candidate.

Dataset

Phase 3 contains 1,089 market-only prompts on the dedicated synthetic Modal volume. The dataset is split into 726 dense prompts and 363 minimal prompts so the report can compare broad coupled variation against a cleaner stripped-down version of the same pairwise structure.

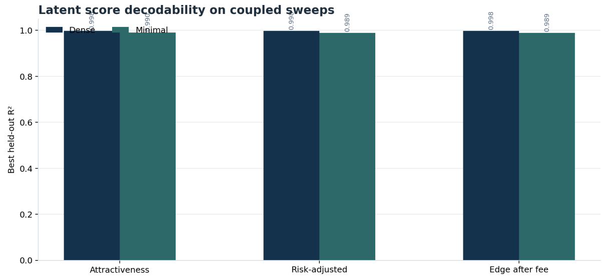


Phase 3 allocates almost all of its scale to coupled-factor sweeps. The minimal slice removes more nuisance structure; the dense slice stress-tests whether the geometry survives richer market backgrounds.

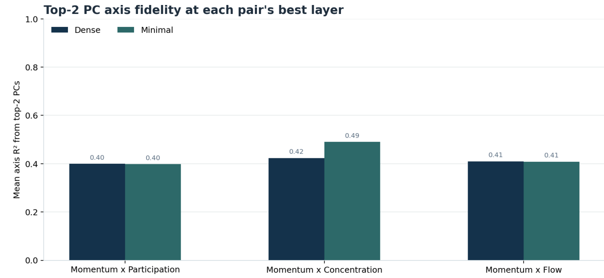
Main Quantitative Findings

Metric	Phase 2 scalar best	Dense coupled	Minimal coupled
Best attractiveness R^2	0.997	0.998	0.990
Best risk-adjusted R^2	0.997	0.998	0.989
Best edge-after-fee R^2	0.997	0.998	0.989
Best scalar / coupled geometry	0.741	0.708	0.693
Best pairwise AUROC	0.997	1.000	0.997

The probe story remains easy: the synthetic latent variables are still almost perfectly recoverable. The important change in this phase is the geometry read, not whether a linear probe can decode the labels.



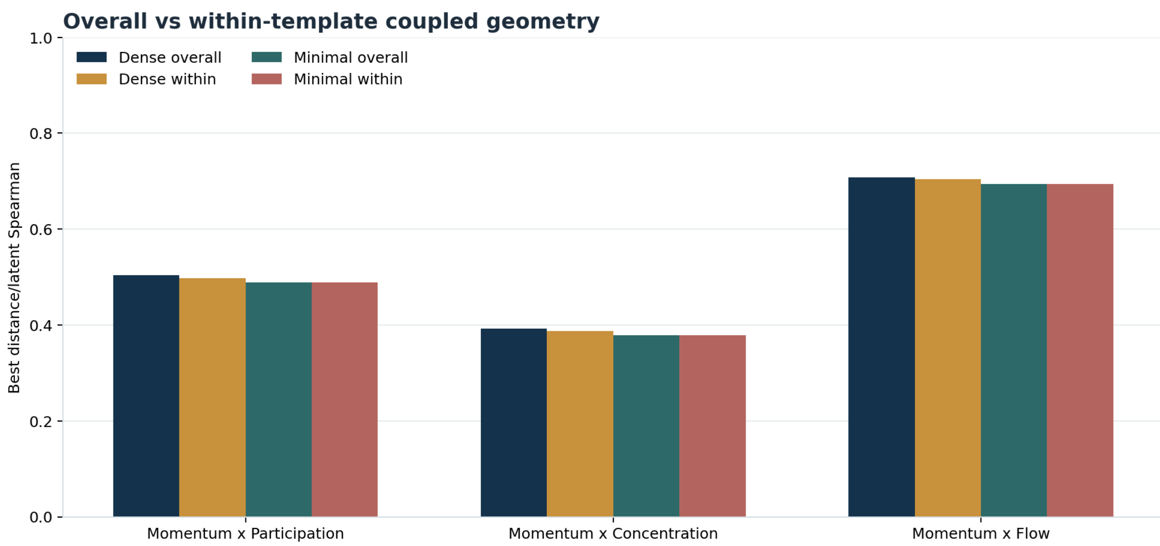
Latent-score regression remains near ceiling in both coupled families, confirming that the synthetic labels are cleanly preserved in row states.



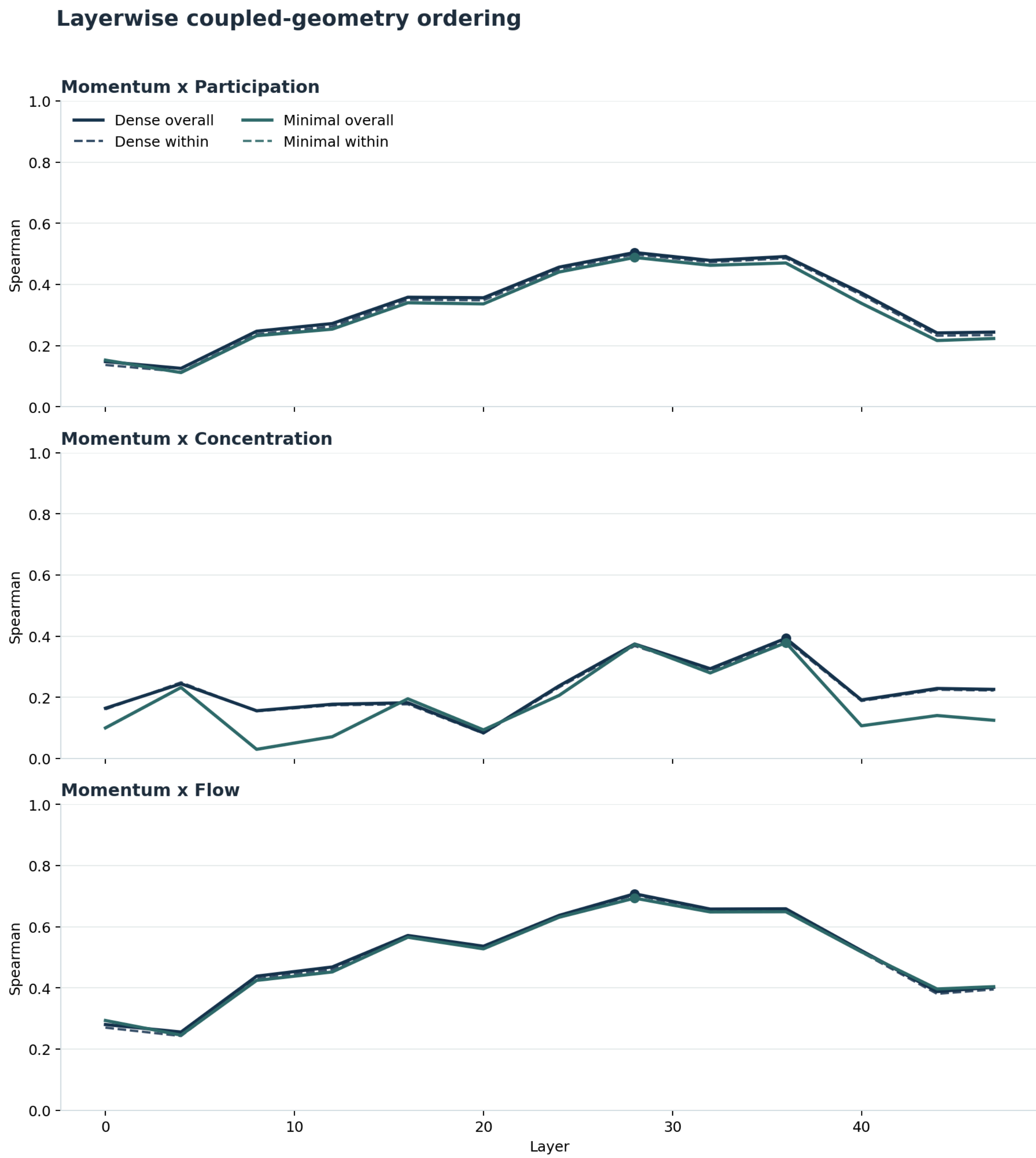
Top-2 PC axis fidelity gives a cleaner read on whether the coupled spaces are behaving like coherent low-dimensional factor systems rather than arbitrary high-dimensional clusters.

Coupled Geometry Read

This is the central stage-transition result. Phase 3 asks whether the right object was never a universal scalar manifold, but a family of small coupled manifolds instead.



Overall versus within-template coupled geometry. Within-template scores test whether the ordering survives after removing the easiest prompt-template differences and reading only the latent geometry within repeated backgrounds.



Layerwise coupled ordering for the three candidate factor pairs. The strongest pair should sustain a broad late-middle layer band rather than a single fragile spike.

Coupled family	Best state	Dense best	Minimal best
`Momentum × Flow`	`row_mean` L28	0.708	0.693
`Momentum × Participation`	`row_mean` L28	0.504	0.488
`Momentum × Concentration`	`row_eos` L36	0.393	0.379

- `pct_5m × net_flow_5m` is the clear lead coupled pair and remains almost unchanged under within-template filtering in the dense slice (0.708 overall vs 0.704 within).

- `pct_5m × unique_traders_5m` is real but materially weaker, suggesting participation behaves more like a secondary modulator than a dominant joint manifold axis.
- `pct_5m × top20_holder_pct` remains weak in both dense and minimal settings, so concentration still does not behave like a clean coupled factor in this synthetic form.
- The strongest coupled layers cluster in the late 20s to mid 30s, which is later than the near-trivial latent decodability and consistent with geometry sharpening after early preservation.

Interpretation

> **SUPPORTED NOW**
>
> At least one coupled market space — momentum × flow — is represented with stable low-dimensional order that survives prompt-template controls rather than disappearing when the synthetic backgrounds vary.

> **NOT SUPPORTED YET**
>
> Coupled geometry is not universally cleaner than the strongest scalar candidate, and we still do not have a dominant joint market manifold that subsumes the Stage 1 scalar search.

Best current interpretation:

- Stage 1 remains necessary: `pct_5m` is still the single cleanest variable we have.
- Stage 2 is now justified: momentum × flow is too stable across dense, minimal, and within-template reads to dismiss as a template artifact.
- Participation does shape the market representation jointly with momentum, but more weakly than flow.
- Concentration is still better treated as a difficult secondary factor than as a clean geometric partner.

What Phase 3 Accomplished

Relative to the staged plan in `MARKET_COUNTING_MANIFOLDS_PLAN.md`, Phase 3 closes the first pass on Stage 2:

- it constructed and captured the first dedicated coupled-factor synthetic dataset
- it evaluated the same pooled row states with coupled-geometry metrics rather than scalar-only metrics
- it compared dense versus minimal prompt families rather than relying on one synthetic setting
- it established whether the coupled story is stronger than the single-factor story from Phase 2

What Is Still Missing

Phase 3 is not yet the full market-manifold result.

- It still studies only three handpicked factor pairs rather than a broader interaction graph.

- It does not yet test settings or portfolio context as twisting operators on the coupled spaces.
- It does not yet include causal interventions on the strongest pairwise geometry candidate.
- It does not yet characterize the higher-dimensional joint market-state space that would correspond to Stage 3 of the plan.

Next Moves

The next step should be targeted rather than broad:

1. deepen the strongest coupled pair with denser local sweeps and repeated backgrounds
2. run causal patching or ablation on the best coupled geometry candidate
3. only then reintroduce settings and portfolio context to test whether those later sections rotate or reweight the coupled space
4. expand to Stage 3 only after the strongest Stage 2 candidate is causally validated

If the strongest coupled candidate fails causal tests, that would be the first real challenge to the new staged hypothesis.

Synthetic Market Phase 3 Coupled Geometry Report — 20 March 2026. 1,089 market-only captures on the dedicated synthetic Modal volume. Reference baseline: Phase 2 scalar geometry report.